The University of Nottingham
Malaysia Campus

SCHOOL OF COMPUTER SCIENCE

A LEVEL 2 MODULE, AUTUMN SEMESTER 2018-2019

OPERATING SYSTEMS AND CONCURRENCY
COMP 2035 (G52OSC)

Time allowed TWO Hours

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*Candidates must **NOT** start writing their answers until told to do so*

Answer THREE questions out of FOUR

Only silent, self-contained calculators with a single line display are permitted in this examination.

Dictionaries are not allowed with one exception. Those whose first language is not English may use a standard translation dictionary to translate between that language and English provided that neither language is the subject of this examination. Subject specific translation dictionaries are not permitted.

No electronic devices capable of storing and retrieving text, including electronic dictionaries, may be used.

DO NOT turn examination paper over until instructed to do so

1. Memory Management (25 Marks)

(a) Explain, with the aid of a diagram, the operation of an address translation process in a paged memory allocation scheme. [7 Marks]

(b) Consider the following segment Table-1.0:

Segment	*Base*	*Length*
0	219	600
1	2300	14
2	90	100
3	1327	580
4	1952	96

Table-1.0

What are the physical addresses for the following logical addresses?

(i) 0, 430

(ii) 1, 10

(iii) 2, 500

(iv) 3, 400

(v) 4, 112

[5 Marks]

(c) What is *thrashing*? How does the system detect it and what the system could do to eliminate it?

[5 Marks]

(d) Given memory partitions of 100K, 500K, 200K, 300K, and 600K (in order), how would each of the First-fit, Best-fit, and Worst-fit algorithms place processes of 212K, 417K, 112K, and 426K (in order)?

[3 Marks]

(e) Which algorithm makes the most efficient use of memory? [1 Mark]

(f) Describe briefly, the second-chance page-replacement policy. [4 Marks]

2. Process Scheduling (25 Marks)

(a) Consider the ready list containing the processes and their burst times as shown in Table 2.0. The processes are assumed to arrive at the same time.

Processes	Burst Time (ms)
P0	350
P1	125
P2	475
P3	250
P4	75

Table 2.0

Based on the above table, draw a timeline to turn it around and calculate the average turn-around and average waiting times for

(i) Non Pre-Emptive, Shortest Job First scheduling,

(ii) Round Robin scheduling with the time quantum = 50 and overhead = 10.

[8 Marks]

(b) With the help of example, briefly describe the Multilevel Feedback Queue scheduling algorithm.

[4 Marks]

(c) Explain the differences between user-level threads and kernel-level threads. List any two advanatages of user level threads.

[4 Marks]

(d) Describe the operations of the three (3) multithreading models. [9 Marks]

3. Deadlock, Device Management and File Management (25 Marks)

(a) Explain what is meant by the term *deadlock*. [2 Marks]

(b) A system is composed of four processes, (p1, p2, p3 and p4), and three types of consumable resources, [R1, R2 and R3]. The number of units available for R1 is 1, R2 is 0 and R3 is 1.

- p1 requests one unit of R1 and one unit of R3.
- p2 produces R1 and R3 and requests one unit of R2.
- P3 requests one unit each of R1 and R3.
- P4 produces R2 and requests one unit of R3.

(i) Draw a consumable resource allocation graph to represent this system state.

[4 Marks]

(ii) Which, if any, of the processes are deadlocked in this state?

[2 Marks]

(c) Briefly describe three ways of allocating storage, and identify the advantages of each.

[9 Marks]

(d) List any five (5) events that could be occurred during DMA when it transfers data from disk to the main memory?

[5 Marks]

(e) Briefly describe the operations of an *elevator disk scheduling* algorithm. List any one limitation of this algorithm.

[3 Marks]

4. Process synchronisation and Mutual Exclusion (25 Marks)

(a) Describe what is meant by *race conditions?* How do they occur?

[4 Marks]

(b) List four conditions that can make a solution to the problem of *race conditions* acceptable.

[4 Marks]

(c) Write an outline implementation for the *strict alternation algorithm*.

[4 Marks]

(d) Critically evaluate the *strict alternation algorithm* with reference to the mutual exclusion problem.

[7 Marks]

(e) What is the meaning of the term *busy waiting*? [3 Marks]

(f) Write any three problems that may arise when disabling the interrupts during the execution of a long critical section.

[3 Marks]